

Techniques to Evasion of Intrusion Prevention Systems (IPS) and Intrusion Detection Systems (IDS)

In the realm of cybersecurity, understanding the techniques used to evade Intrusion Prevention Systems (IPS) and Intrusion Detection Systems (IDS) is crucial for both attackers and defenders. This document explores various methods employed by malicious actors to bypass these security measures, highlighting the importance of continuous improvement in security protocols and technologies.

1. Packet Fragmentation

One common technique is packet fragmentation, where an attacker breaks down malicious payloads into smaller packets. By doing so, they can evade detection, as the IPS/IDS may not reassemble the packets correctly or may overlook the fragmented packets altogether.

2. Encryption

Using encryption to obfuscate the payload is another effective method. Attackers can encrypt their malicious traffic, making it difficult for IPS/IDS to analyze the content. Even if the traffic is detected, the encrypted nature may prevent the system from identifying it as a threat.

3. Polymorphic Code

Polymorphic code changes its appearance each time it is executed. This technique allows malware to evade signature-based detection methods employed by many IPS/IDS systems, as the changing code can bypass static analysis.

4. Tunneling

Tunneling involves encapsulating malicious traffic within a legitimate protocol, such as HTTP or DNS. By disguising the traffic, attackers can bypass detection mechanisms that are not configured to inspect the encapsulated data.

5. Use of Legitimate Tools

Attackers often leverage legitimate tools and services to carry out their attacks. By using tools that are commonly accepted and trusted, they can blend in with normal traffic, making it challenging for IPS/IDS to distinguish between malicious and benign activities.

6. Timing Attacks

Timing attacks exploit the timing of packet transmissions to evade detection. By sending packets at intervals that mimic normal traffic patterns, attackers can avoid triggering alerts that would typically be raised by unusual traffic spikes.

7. Anomaly Avoidance

Attackers may study the behavior of the target network and adjust their tactics to avoid triggering anomalies that would be flagged by an IDS. By understanding the baseline of normal activity, they can craft their attacks to blend in seamlessly.

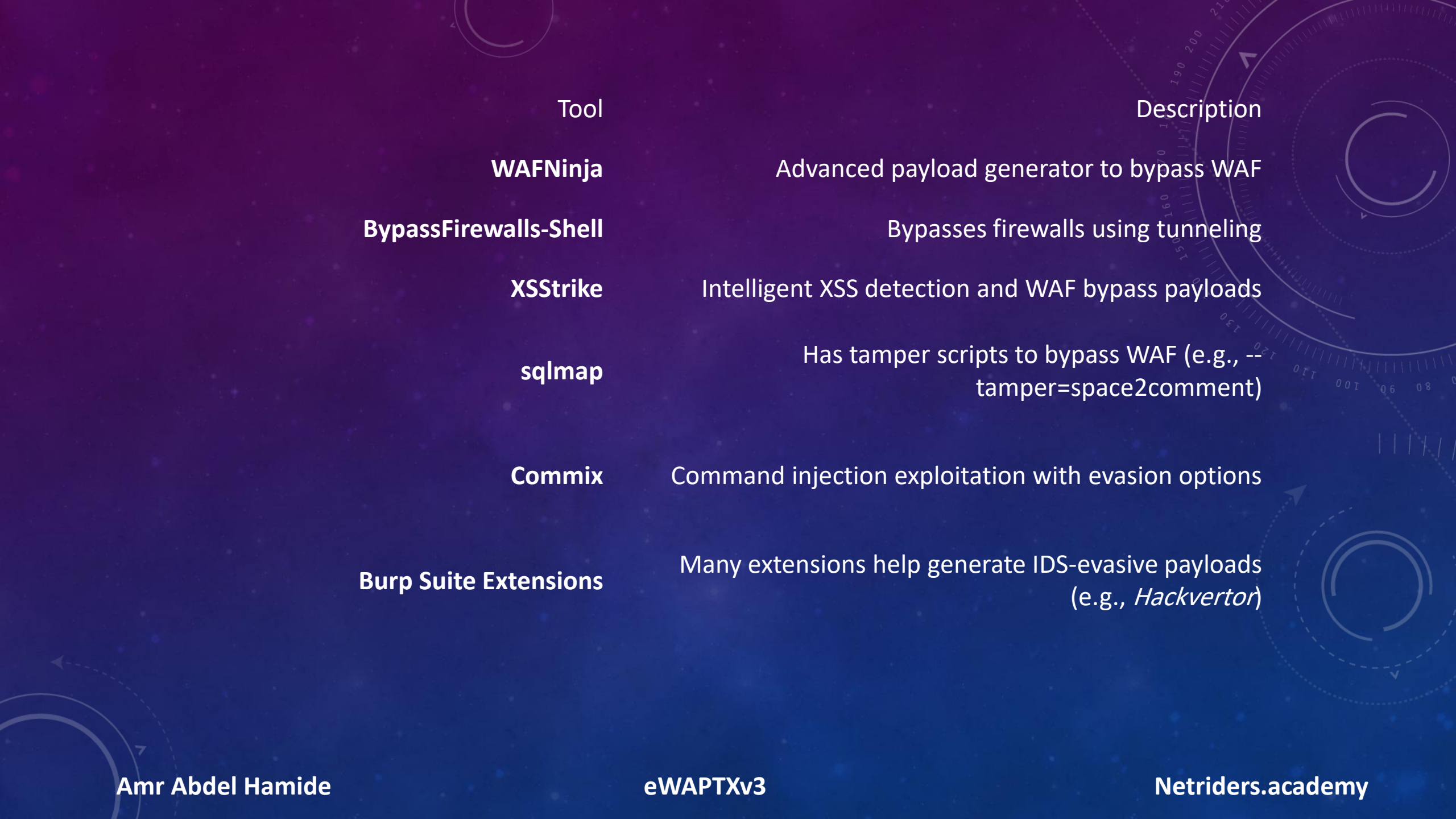
8. Use of Decoy Traffic

Generating decoy traffic can confuse IPS/IDS systems. By flooding the network with benign traffic, attackers can mask their malicious activities, making it harder for detection systems to identify real threats.

Tools for Bypassing IDS/IPS (Organized by Category)

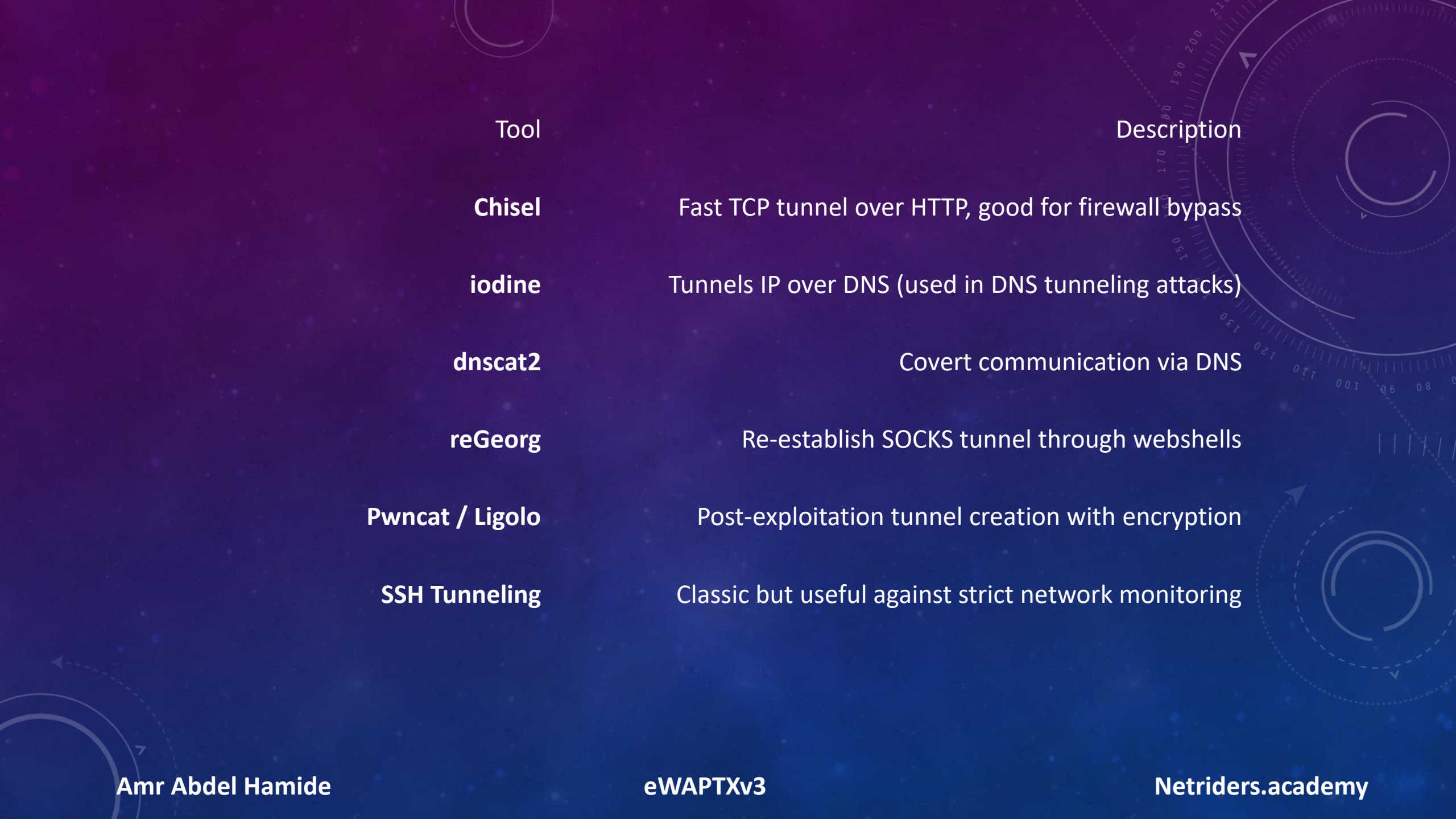
<u>Tool</u>	Description
Veil	Obfuscates payloads to evade AV/IDS/IPS
Shellter	Injects shellcode into PE files, with evasion
msfvenom	Supports encoding payloads using -e
obfuscation.io	Online JS/PHP/HTML obfuscator
Hyperion	Runtime crypter for Windows executables
TheFatRat	Generates and obfuscates FUD backdoors

Web Attack Bypass Tools (WAF/IDS)



Tool	Description
WAFNinja	Advanced payload generator to bypass WAF
BypassFirewalls-Shell	Bypasses firewalls using tunneling
XSSStrike	Intelligent XSS detection and WAF bypass payloads
sqlmap	Has tamper scripts to bypass WAF (e.g., --tamper=space2comment)
Commix	Command injection exploitation with evasion options
Burp Suite Extensions	Many extensions help generate IDS-evasive payloads (e.g., <i>Hackvector</i>)

Custom Protocol/Encryption Tunneling Tools



Tool	Description
Chisel	Fast TCP tunnel over HTTP, good for firewall bypass
iodine	Tunnels IP over DNS (used in DNS tunneling attacks)
dnscat2	Covert communication via DNS
reGeorg	Re-establish SOCKS tunnel through webshells
Pwncat / Ligolo	Post-exploitation tunnel creation with encryption
SSH Tunneling	Classic but useful against strict network monitoring